

Q1 B: Derive the equation of the osculating plane for a curve given by $\mathbf{r}(t)$. Provide a geometrical interpretation of the osculating plane.

Setup and Definitions

Let $\mathbf{r}(t)$ be a smooth curve in $\mathbb{R}^3$, parameterized by t , and assume $\mathbf{r}'(t) \neq \mathbf{0}$ (the curve is regular). The **osculating plane** at a point $t = t_0$ is the plane that best approximates the curve locally — it contains the tangent line and the direction in which the curve is bending.

Step 1: Unit Tangent Vector

The unit tangent vector is:

$$\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|}$$

Step 2: Principal Normal Vector

Since $|\mathbf{T}(t)| = 1$, differentiating gives $\mathbf{T}(t) \cdot \mathbf{T}'(t) = 0$, so $\mathbf{T}'(t) \perp \mathbf{T}(t)$.

The principal unit normal vector (defined when $\mathbf{T}'(t) \neq \mathbf{0}$) is:

$$\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{|\mathbf{T}'(t)|}$$

The condition $\mathbf{T}'(t_0) \neq \mathbf{0}$ is required for the osculating plane to be well-defined (it fails, for instance, at inflection points of straight lines).

Step 3: Binormal Vector

The **binormal vector** is the unit vector perpendicular to the osculating plane, defined as:

$$\mathbf{B}(t) = \mathbf{T}(t) \times \mathbf{N}(t)$$

Since $\mathbf{T}$ and $\mathbf{N}$ are orthonormal, $|\mathbf{B}| = 1$ and $\mathbf{B} \perp \mathbf{T}$, $\mathbf{B} \perp \mathbf{N}$.

Step 4: Equation of the Osculating Plane

The osculating plane at $t = t_0$ passes through $\mathbf{r}(t_0)$ and has $\mathbf{B}(t_0)$ as its normal. A point $\mathbf{r}$ lies in this plane if and only if:

$$\boxed{\mathbf{B}(t_0) \cdot (\mathbf{r} - \mathbf{r}(t_0)) = 0}$$

Step 5: Practical Form via $\mathbf{r}'$ and $\mathbf{r}''$

In practice, $\mathbf{B}$ need not be fully normalized. Since $\mathbf{T} = \frac{\mathbf{r}'}{|\mathbf{r}'|}$, one can show:

$$\mathbf{r}'(t) \times \mathbf{r}''(t) \parallel \mathbf{T}(t) \times \mathbf{N}(t) = \mathbf{B}(t)$$

because the cross product $\mathbf{r}' \times \mathbf{r}''$ differs from $\mathbf{B}$ only by a positive scalar factor. Therefore the osculating plane equation can equivalently be written as:

$$\boxed{(\mathbf{r}'(t_0) \times \mathbf{r}''(t_0)) \cdot (\mathbf{r} - \mathbf{r}(t_0)) = 0}$$

Proof that $\mathbf{r}' \times \mathbf{r}'' \parallel \mathbf{B}$

Let s denote arc length so that $\mathbf{r}' = |\mathbf{r}'| \mathbf{T}$. Differentiating:

$$\mathbf{r}'' = |\mathbf{r}'|' \mathbf{T} + |\mathbf{r}'| \mathbf{T}' = |\mathbf{r}'|' \mathbf{T} + |\mathbf{r}'|^2 \kappa \mathbf{N}$$

where $\kappa = |\mathbf{T}'|/|\mathbf{r}'|$ is the curvature. Then:

$$\mathbf{r}' \times \mathbf{r}'' = |\mathbf{r}'| \mathbf{T} \times (|\mathbf{r}'|' \mathbf{T} + |\mathbf{r}'|^2 \kappa \mathbf{N}) = |\mathbf{r}'|^3 \kappa (\mathbf{T} \times \mathbf{N}) = |\mathbf{r}'|^3 \kappa \mathbf{B}$$

since $\mathbf{T} \times \mathbf{T} = \mathbf{0}$. Thus $\mathbf{r}' \times \mathbf{r}''$ is a positive multiple of $\mathbf{B}$, confirming the equivalence. $\square$

Geometric Interpretation

- The osculating plane “*kisses*” (from Latin *osculari*) the curve at $\mathbf{r}(t_0)$, providing the best planar approximation to second order.
- It contains both $\mathbf{T}(t_0)$ (the direction of motion) and $\mathbf{N}(t_0)$ (the direction of curvature), capturing the curve’s instantaneous direction *and* bending.
- The binormal $\mathbf{B}(t_0)$ is perpendicular to the osculating plane and measures how much the curve twists out of the plane (governed by torsion τ).
- Together $\{\mathbf{T}, \mathbf{N}, \mathbf{B}\}$ form the **Frenet–Serret frame**, an orthonormal basis that moves along the curve.

Summary

$$\text{Osculating plane (abstract form): } \mathbf{B}(t_0) \cdot (\mathbf{r} - \mathbf{r}(t_0)) = 0$$

$$\text{Osculating plane (practical form): } (\mathbf{r}'(t_0) \times \mathbf{r}''(t_0)) \cdot (\mathbf{r} - \mathbf{r}(t_0)) = 0$$